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Title: Vision-Driven Alertness - A Hybrid Framework Combining Deep Learning and Image Processing for Real-Time Drowsiness Detection and Road Safety Enhancement

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Abstract: Drowsy driving significantly endangers road safety, often causing accidents. This paper presents a real-time drowsiness detection system using facial landmarks and image processing. The system uses Eye Aspect Ratio (EAR) and Mouth Aspect Ratio (MAR) with Convolutional Neural Networks (CNNs) and Haar cascades for facial feature detection, achieving 94 % accuracy in eye closure detection and 96 % in yawning detection. The system demonstrates robust performance across diverse lighting conditions and driver orientations, making it suitable for real-world deployment. Future enhancements include integration with vehicle systems, adaptive alerts, and incorporating physiological signals for comprehensive monitoring. This system has the potential to significantly reduce fatigue-related accidents, enhancing road safety.

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